PHYSIOLOGY · MCQ PRACTICE

Swallowing (Deglutition) — MCQ Practice

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GIT BLOCK

Physiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Questions

1. Which stage of swallowing is under voluntary control? EASY

A. The pharyngeal stage
B. The esophageal stage
C. The oral (voluntary) stage
D. None — all three stages are involuntary
E. All three stages are voluntary

2. Where is the deglutition (swallowing) centre located? EASY

A. Cerebral cortex
B. Hypothalamus
C. Medulla oblongata and lower pons
D. Cerebellum
E. Midbrain

3. Which structure closes off the larynx during the pharyngeal stage of swallowing to prevent aspiration? EASY

A. Soft palate
B. Uvula
C. Epiglottis
D. Upper esophageal sphincter
E. Cricoid cartilage

4. Approximately how long does it take a solid or semisolid bolus to transit the esophagus? EASY

A. ~1 second
B. 4–8 seconds
C. 30–60 seconds
D. 2–5 minutes
E. Less than 0.5 seconds

5. A patient with a brainstem stroke affecting the medulla develops difficulty initiating the involuntary phase of swallowing after the bolus reaches the oropharynx, though he can still voluntarily push food to the back of his mouth. Which stage is most impaired? MEDIUM

A. Oral (voluntary) stage
B. Pharyngeal stage
C. Esophageal stage only
D. All three stages equally, since the medulla controls all swallowing
E. None — this describes normal swallowing

6. A patient regurgitates food and liquid through the nose while swallowing. Which mechanism has most likely failed? MEDIUM

A. Epiglottis closure over the larynx
B. Upward movement of the soft palate and uvula sealing the nasopharynx
C. Relaxation of the lower esophageal sphincter
D. Circular muscle contraction above the bolus
E. Peristaltic wave initiation

7. During esophageal peristalsis, which combination of muscle activity correctly propels the bolus forward? MEDIUM

A. Circular muscle above the bolus contracts; longitudinal muscle below it contracts to widen that segment
B. Longitudinal muscle above the bolus contracts; circular muscle below it contracts
C. Both circular and longitudinal muscle contract simultaneously along the entire esophagus
D. Only circular muscle is involved in peristalsis; longitudinal muscle plays no role
E. Only longitudinal muscle is involved; circular muscle plays no role

8. A patient with achalasia has failure of lower esophageal sphincter relaxation. At which point in the swallowing sequence would food most likely become held up? MEDIUM

A. Entry into the oropharynx
B. Closure of the nasopharynx
C. Transit through the pharynx
D. The junction where the esophagus meets the stomach
E. Opening of the upper esophageal sphincter

9. What is the physiological purpose of esophageal gland mucus during swallowing? MEDIUM

A. It activates pepsinogen
B. It closes the nasopharynx
C. It lubricates the bolus through the esophagus and reduces friction, taking over from saliva
D. It triggers the pharyngeal stage
E. It relaxes the lower esophageal sphincter

10. A patient swallows a sip of water followed by a bite of solid bread. Which statement about their relative transit times through the esophagus is correct? MEDIUM

A. The water and bread transit at the same rate, since peristalsis is a fixed-speed wave
B. The water transits in about 1 second, much faster than the 4–8 seconds needed for the bread
C. The bread transits faster because it is denser
D. Neither is affected by consistency; only bolus volume matters
E. Liquids always take longer because they lack the compact shape needed for peristaltic gripping

11. Which sequence correctly orders the events of the pharyngeal stage after the bolus stimulates oropharyngeal receptors? MEDIUM

A. Epiglottis closes → soft palate rises → impulses reach the deglutition centre → bolus enters esophagus
B. Impulses reach the deglutition centre → soft palate/uvula seal the nasopharynx → epiglottis closes the larynx → bolus transits into the esophagus
C. Bolus enters esophagus → deglutition centre fires → soft palate rises
D. Upper esophageal sphincter relaxes first, before any brainstem involvement
E. Longitudinal esophageal muscle contracts before the deglutition centre is activated

12. A surgeon needs to identify the structure whose relaxation allows the bolus to leave the pharynx and enter the esophagus. Which structure is this? MEDIUM

A. Lower esophageal sphincter
B. Pyloric sphincter
C. Upper esophageal sphincter
D. Ileocecal valve
E. Cardiac sphincter of the stomach

13. A patient with bulbar palsy (lower motor neuron dysfunction affecting brainstem motor nuclei) is at particular risk for aspiration pneumonia. Which physiological explanation best accounts for this? MEDIUM

A. The esophageal peristaltic reflex is entirely independent of the brainstem and would be unaffected, so aspiration risk comes only from the oral stage

B. Motor output from the medullary deglutition centre governing pharyngeal protective reflexes (soft palate elevation, epiglottis closure) is impaired

C. Bulbar palsy only affects voluntary tongue movement, with no bearing on airway protection

D. The lower esophageal sphincter fails to relax, causing reflux into the airway

E. Sympathetic outflow to the esophagus is lost, halting peristalsis entirely

14. A patient with a peptic stricture near the gastroesophageal junction reports that solid food "sticks" a few seconds after swallowing, matching the expected transit time for solids through a healthy esophagus. What does this timing suggest about where the obstruction most likely sits? MEDIUM

A. At the level of the oropharynx, since 4–8 seconds is too fast for a distal problem

B. Near the distal esophagus/gastroesophageal junction, consistent with normal solid transit time before the bolus meets the obstruction

C. In the nasopharynx, since sticking sensations always localize there

D. The timing described is inconsistent with any esophageal pathology

E. At the upper esophageal sphincter exclusively, since that is the fastest transit point

15. A patient has a lesion isolated to the cerebral cortex (voluntary motor areas) with an intact brainstem. Which outcome is most likely for their swallowing? HARD

A. Complete inability to swallow at all, since the cortex controls all three stages

B. Difficulty voluntarily initiating the oral stage, but the pharyngeal and esophageal stages will proceed normally once triggered

C. Loss of epiglottis closure specifically

D. Normal swallowing in every respect, since the cortex plays no role in deglutition

E. Loss of lower esophageal sphincter relaxation

16. In a patient with a nasogastric tube in place, esophageal peristalsis is observed to still occur around the tube when a bolus is swallowed. What does this best demonstrate about esophageal motility? HARD

A. Esophageal peristalsis is entirely dependent on an empty, unobstructed lumen

B. The reflex nature of esophageal peristalsis (mediated by sequential circular/longitudinal muscle contraction) persists even with a foreign body present

C. The presence of a tube abolishes the pharyngeal stage

D. Peristalsis in this case is purely due to gravity, not muscular contraction

E. This indicates failure of the deglutition centre

17. Two patients present with dysphagia. Patient X has difficulty only with liquids, coughing immediately after sipping water, with normal swallowing of solids. Patient Y has the reverse — solids stick but liquids pass easily. Which explanation best fits this pattern? HARD

A. Patient X likely has a pharyngeal-stage/airway-protection problem (impaired epiglottis or soft palate closure, which fast liquids expose more readily); Patient Y likely has a mechanical narrowing or peristaltic weakness better tolerated by liquids

B. Both patients have identical underlying pathology

C. Patient X has an esophageal stricture; Patient Y has a pharyngeal reflex problem

D. Neither presentation relates to the three-stage model of swallowing

E. Patient Y's pattern indicates loss of taste rather than a motor problem

18. A study measures intraluminal esophageal pressure during swallowing and finds a high-pressure zone travelling aborally (toward the stomach) that always stays just ahead of a lower-pressure zone. This pattern is best explained by: HARD

A. Simultaneous, non-sequential contraction of the entire esophagus

B. The circular muscle contraction wave (high pressure) advancing just ahead of the relaxed, receiving segment created by longitudinal contraction (relatively lower pressure)

C. Retrograde peristalsis pushing the bolus backward

D. Failure of the lower esophageal sphincter to open

E. Random, uncoordinated esophageal spasm

19. A drug that selectively paralyzes striated (skeletal) muscle, but not smooth muscle, is administered. Which portion of swallowing would be most affected? HARD

A. Only the lower esophageal sphincter, since it is skeletal muscle

B. The oral and pharyngeal stages and the proximal (striated-muscle) esophagus, while the distal smooth-muscle esophagus and LES continue functioning

C. None of the swallowing stages, since deglutition uses only smooth muscle

D. Only the distal esophagus, since it is entirely skeletal muscle

E. All three stages equally, since swallowing muscle type is uniform throughout

20. A patient has a vagus nerve lesion that spares afferent fibers from the oropharynx but disrupts efferent output to the pharyngeal musculature and esophagus. Predict the most likely combined effect. HARD

A. Sensory triggering of the swallow reflex is lost entirely, with intact motor output

B. The deglutition centre can still be triggered by oropharyngeal sensation, but the coordinated motor response (pharyngeal constriction, sphincter relaxation, esophageal peristalsis) will be impaired or absent

C. Only taste is lost, with no effect on swallowing mechanics

D. Swallowing will be entirely normal because the vagus plays no efferent role in deglutition

E. Only the voluntary oral stage will be affected

✓ Answer key

Q	Answer	Difficulty	Why
1	C	EASY	The oral/voluntary stage is initiated consciously; the pharyngeal and esophageal stages are involuntary.
2	C	EASY	The deglutition centre that governs the pharyngeal stage sits in the medulla oblongata and lower pons.
3	C	EASY	The epiglottis folds over the laryngeal opening, protecting the airway.
4	B	EASY	Solid/semisolid boluses take about 4–8 seconds to transit the esophagus, versus ~1 second for liquids.
5	B	MEDIUM	The pharyngeal stage depends on the medullary deglutition centre; the voluntary oral stage (cortex-driven) is preserved.
6	B	MEDIUM	Failure of soft palate/uvula elevation allows food or liquid to escape into the nasopharynx and nose.
7	A	MEDIUM	Circular muscle contracts above the bolus to squeeze it forward while longitudinal muscle below contracts to shorten and widen the receiving segment.
8	D	MEDIUM	Failure of LES relaxation prevents the bolus from passing from the esophagus into the stomach.
9	C	MEDIUM	Once past the pharynx, esophageal gland mucus (not saliva) lubricates the bolus.
10	B	MEDIUM	Very soft food/liquid transits in ~1 second; solid/semisolid takes 4–8 seconds.
11	B	MEDIUM	Sensory impulses travel to the medullary deglutition centre first, which then coordinates nasopharyngeal closure, laryngeal closure, and esophageal entry in sequence.
12	C	MEDIUM	The upper esophageal sphincter relaxes to allow the bolus to pass from the pharynx into the esophagus.
13	B	MEDIUM	The pharyngeal stage protective reflexes depend on motor output from brainstem nuclei coordinated by the deglutition centre; their impairment in bulbar palsy directly increases aspiration risk.
14	B	MEDIUM	Since normal solid transit through the esophagus takes about 4–8 seconds, a sticking sensation appearing after that interval is consistent with an obstruction near the distal esophagus/GE junction, which the bolus reaches only after traversing that normal transit time.
15	B	HARD	Only the oral stage requires voluntary cortical initiation; the pharyngeal and esophageal stages are reflexive and brainstem/myenteric-plexus driven, so they remain intact if triggered.

16	**B**	HARD	Esophageal peristalsis is a coordinated neuromuscular reflex triggered by bolus-induced distension and continues functioning even with a coexisting foreign body, illustrating the robustness of the sequential circular/longitudinal contraction mechanism.
17	**A**	HARD	Liquid-predominant dysphagia with coughing suggests a pharyngeal/airway-protection defect (fast-moving liquids more easily breach an incompletely closed airway), while solid-predominant dysphagia suggests mechanical obstruction or weak peristalsis, since liquids can still get past a narrowing that solids cannot.
18	**B**	HARD	This is the normal peristaltic pressure signature: a contraction (high-pressure) wave in circular muscle propels the bolus, staying just ahead of the receiving (lower-pressure, longitudinally shortened) segment.
19	**B**	HARD	The oral cavity, pharynx, and upper esophagus are striated (skeletal) muscle under somatic motor control, while the distal esophagus and LES are smooth muscle; selective skeletal muscle paralysis would impair the former while sparing the latter.
20	**B**	HARD	The vagus provides key motor (efferent) output for pharyngeal and esophageal stages; disrupting efferent function while sparing afferent sensory input means the swallow reflex can still be triggered but its execution will fail.